

Global Zombie Outbreak Modelling: Survival Strategy Analysis

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Introduction

This memo explains the mathematical modelling of a global zombie outbreak and evaluates how different intervention strategies affect the long-term survival of the human population. Two primary strategies are available: develop a vaccine to reduce human susceptibility to the zombie-disease or provide weapons and training to enable humans to fight back effectively. Given resource constraints, only one strategy can be pursued. This memo examines the projected outcomes of each approach and determines which offers a better chance of human survival.

The Model

The outbreak dynamics follow a system of ODEs, as given in [1], with three populations: humans $H(t)$, zombies $Z(t)$, and removed zombies $R(t)$.

$$\begin{aligned}H'(t) &= -\beta H(t)Z(t) \\Z'(t) &= \beta H(t)Z(t) + \zeta R(t) - \alpha H(t)Z(t) \\R'(t) &= \alpha H(t)Z(t) - \zeta R(t)\end{aligned}$$

where α is human removal effectiveness, β is zombie infection rate, and ζ is removal reversion rate. For this analysis, it is assumed that $\zeta = 0$ since no cases of removal reversion have been reported yet. For this analysis, the base model parameters are agreed to be $\alpha = 0.5$ and $\beta = 0.6$.

Model Limitations

The model assumes no natural human deaths or births, which will not be accurate over a one-year horizon. More critically, the infection and removal rates are constant across space and time, meaning all humans are equally vulnerable and all zombies equally effective. The model also treats humans and zombies as simplified groups, ignoring age groups, immunity variations etc. These assumptions simplify the model but they may hide important dynamics of the actual evolution of the zombie-disease.

Intervention Modelling

The base model parameters are subject to large uncertainties of approximately $\pm 25\%$. The two strategies modify the base parameters as follows.

- **Strategy 1: Training and Arming:** Equipping and training humans to fight zombies more effectively increases the removal rate by 50%, giving $\alpha_{\text{new}} = 1.5\alpha$.
- **Strategy 2: Vaccination:** A vaccine reduces human susceptibility to infection by 50%, giving $\beta_{\text{new}} = 0.5\beta$.

For each strategy, we analyse baseline scenarios using the given parameter values and worst-case scenarios where parameter uncertainties are unfavourable: α is reduced by 25% and β is increased by 25%. This range captures the plausible outcomes given our data limitations.

Scenarios

We simulate a population of 1000 humans over one year under all strategy and parameter combinations. Table 1 summaries final populations at $t = 1$ year.

Table 1: Population outcomes by strategy and parameter scenario.

Scenario	Humans	Zombies	Removed
Baseline (No intervention)	0	177	833
Baseline + Strategy 1 (Arming)	960	0	50
Baseline + Strategy 2 (Vaccination)	985	0	25
Worst-case + Strategy 1 (Arming)	0	260	750
Worst-case + Strategy 2(Vaccination)	24	10	976

Without intervention, humanity is eliminated entirely. Both strategies succeed under baseline conditions, with Strategy 2 preserving slightly more humans (985 vs 960). However, under worst-case parameter uncertainty, Strategy 1 fails completely while Strategy 2 preserves a small surviving population of 24 humans.

Detailed population dynamics are visualised in Figure 1. In the worst-case scenario, Strategy 1 fails within months and humans reach zero population, i.e. global human extinction. Strategy 2 keeps a small population alive towards the end of the year.

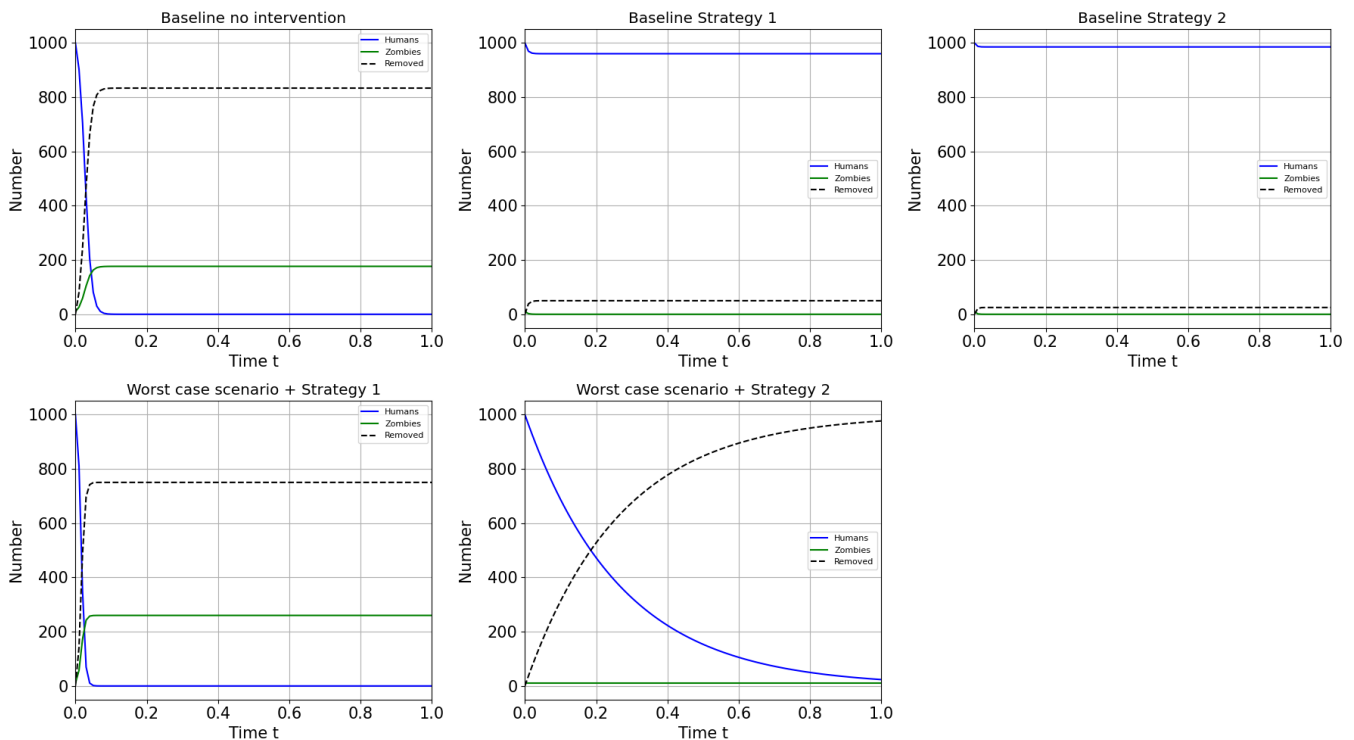


Figure 1: Evolution of population in baseline and worst-case scenarios.

Recommendation

Strategy 2: Vaccination should be adopted. It dominates across all scenarios. At baseline, it preserves 985 humans compared to 960 preserved by training and arming. At worst-case, vaccination preserves 24 humans while training and arming collapses to zero in a short period of time. Strategy 2 wins in every plausible condition.

References

- [1] Griffiths, D. F., & Higham, D. J. (2010). *Numerical Methods for Ordinary Differential Equations*. Springer-Verlag London Limited.